

Structural Controls on Gold Mineralisation

Kalgoorlie Terrane, Eastern Goldfields Province, Western Australia

A regional-scale GIS study integrating geological, structural and mineral-occurrence datasets in QGIS to examine the spatial relationship between recorded gold occurrences and mapped fault systems. Because the Eastern Goldfields is one of the world's best-documented orogenic gold provinces, this project is presented as a **reproducibility and methodology-validation exercise**: its purpose is to demonstrate a complete, defensible regional prospectivity workflow — not to claim a new geological result.

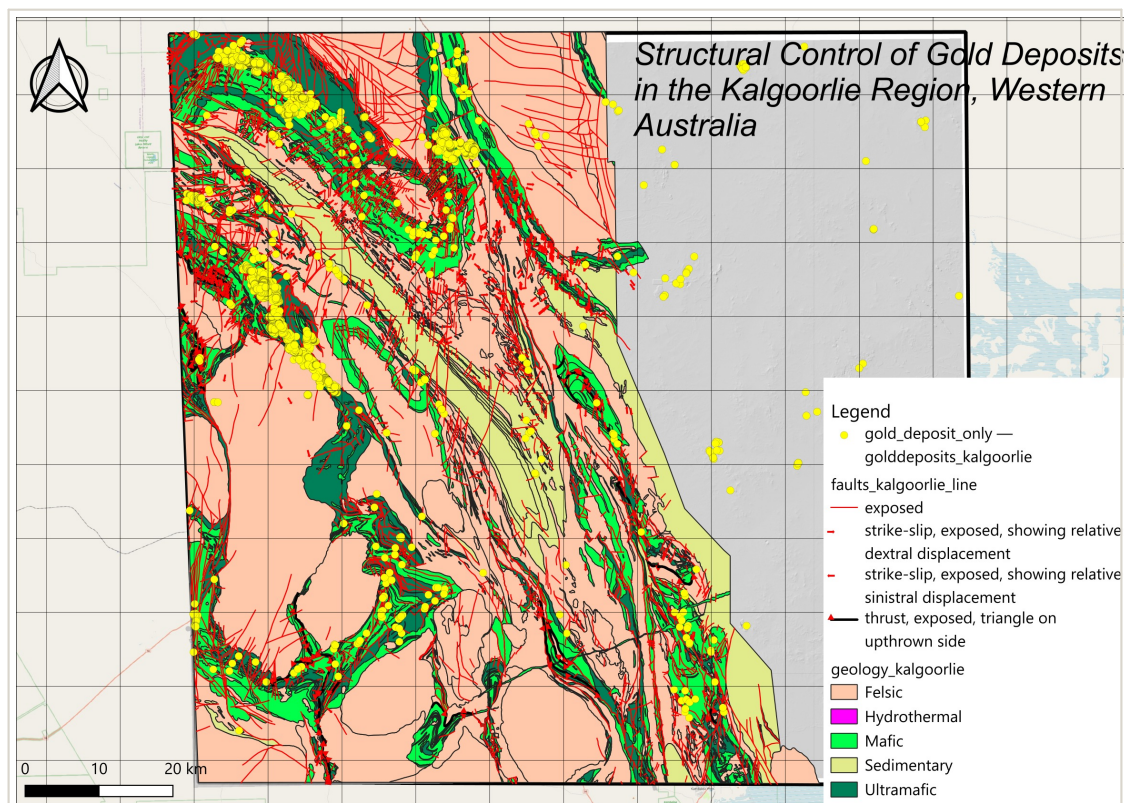


Figure 1. Composite thematic map of the Kalgoorlie district: simplified interpreted geology (mafic, ultramafic, felsic, sedimentary, hydrothermal), mapped fault systems (exposed, strike-slip dextral/sinistral, thrust) and recorded gold occurrences. Projection: GDA94 / MGA Zone 51 (EPSG:28351). Produced in QGIS.

01 Objective & Framing

The Kalgoorlie district hosts world-class orogenic gold mineralisation whose structural control is long-established in the literature. Rather than treat this as an open question, the project uses it as a known-answer test case to validate an analytical workflow. The objectives were to:

- Reproduce the established regional observation that gold occurrences are spatially associated with major fault systems and favourable greenstone lithologies;
- Demonstrate an end-to-end GIS pipeline — coordinate-system management, raster terrain analysis, multi-source data integration, spatial querying and proximity analysis, and publication-quality cartography;
- Document the workflow so it is transparent, reproducible and extensible to other terranes.

02 Study Area

The area of interest covers the Kalgoorlie district within the Eastern Goldfields Superterrane of the Yilgarn Craton. Regionally, gold mineralisation is concentrated along north-northwest–trending structural corridors and shows a well-documented affinity for mafic and ultramafic greenstone units, consistent with a fault-controlled, fluid-pathway

model of orogenic gold formation. These established relationships provide the reference framework against which the GIS results are checked.

03 Data Sources

Dataset	Type	Content / Derived products
Digital Elevation Model	Raster	Terrain base; hillshade, slope and aspect derivatives. Projected in GDA94 / MGA Zone 51 (EPSG:28351).
Interpreted structure (1:100k)	Polyline	Faults, strike-slip faults, thrust faults and structural lineaments.
Interpreted geology (1:100k)	Polygon	Bedrock geology simplified into mafic, ultramafic, felsic, sedimentary and hydrothermal groups.
Mineral occurrences	Point	Gold-related records extracted from the regional mineral-occurrence database.

All layers derive from publicly available Geological Survey of Western Australia (GSWA) regional datasets; no proprietary or restricted data were used.

04 Methodology

All layers were imported into QGIS, reprojected to a common coordinate system (EPSG:28351) and clipped to an area of interest defined by the DEM extent. A proximity buffer was generated around the mapped fault network, gold occurrences were extracted from the mineral-occurrence database, and a spatial intersection was performed to quantify how many occurrences fall within the buffered zone. Occurrences and structures were then overlain on the simplified geology to assess lithological association.



05 Results

17,521 recorded gold-related occurrences in the AOI	16,664 occurrences within the fault proximity buffer	95.1% of occurrences within 5 km of a mapped fault
---	--	--

Spatially, gold occurrences cluster along the north-northwest structural corridors and are preferentially associated with mafic and ultramafic units (Figure 1). This pattern is consistent with the established structural-control model for the district. The high buffer-association figure, however, should be read with the interpretive caveat set out in Section 6 below.

06 Interpretation & Methodological Considerations

Why the 5 km figure alone does not prove structural control. The Eastern Goldfields is a structurally dense terrane: at a 5 km buffer distance the buffered zone occupies the large majority of the AOI. In that setting a ~95% association is close to what a spatially random point pattern would produce, so on its own the statistic is not diagnostic. Establishing genuine structural control requires comparing the observed association against an explicit null expectation.

A defensible regional proximity analysis therefore proceeds as follows:

- **Area normalisation (enrichment factor).** Compare the proportion of occurrences inside the buffer against the proportion of AOI area the buffer occupies. An enrichment factor (observed ÷ expected) greater than 1 indicates a real association; a value near 1 indicates the pattern is explained by buffer coverage alone.
- **Multi-distance buffering.** Repeat the proximity test at a range of shorter distances (for example 250 m, 500 m, 1 km, 2 km) to characterise distance-decay. A true structural control produces enrichment that strengthens monotonically toward the faults.
- **Null-model comparison.** Compare the observed occurrence-to-fault distance distribution against distributions from randomised (Monte Carlo) point sets to test statistical significance.

The present study reports the 5 km proximity result as executed and identifies these refinements as the correct next step. Framing the analysis this way — reporting the result together with its limitations and the path to a rigorous prospectivity claim — reflects standard practice in regional mineral-prospectivity assessment.

07 Recommended Extensions

- Weights-of-Evidence or logistic-regression prospectivity modelling to combine structural, lithological and proximity evidence into a single, statistically weighted map;
- Fault-intersection and structural-complexity density mapping (e.g. kernel density of fault nodes), since dilational jogs and intersections commonly localise orogenic gold;
- Explicit treatment of lithological contacts and greenstone-belt margins as separate evidence layers;
- Filtering the occurrence dataset to distinguish significant deposits from minor prospects and historical workings before quantitative analysis.

08 GIS Capabilities Demonstrated

Capability	Applied in this project
Coordinate-system management	Reprojection and consistent handling of GDA94 / MGA Zone 51 (EPSG:28351).
Raster terrain analysis	Hillshade, slope and aspect generation from DEM data.
Multi-source data integration	Combining raster, polyline and polygon layers into one analytical framework.
Spatial querying & proximity	Buffering, clipping and spatial-intersection selection.
Structural interpretation handling	Managing classified fault types and lineament data.
Cartographic production	Publication-style output with legend, scale bar and north arrow.

09 Data, Reproducibility & Notes

- **Software:** QGIS (open-source), using standard vector-overlay and raster-analysis tools.
- **Projection:** GDA94 / MGA Zone 51 (EPSG:28351) throughout.
- **Data:** publicly available GSWA regional geological, structural and mineral-occurrence datasets.
- **Occurrence counts** reflect all gold-related records in the source database, including prospects and historical workings; deposit-level filtering is noted as a refinement in Section 7.
- **Status:** self-directed demonstration project undertaken to illustrate a regional GIS prospectivity workflow.